

H2 Paper 3 Answers

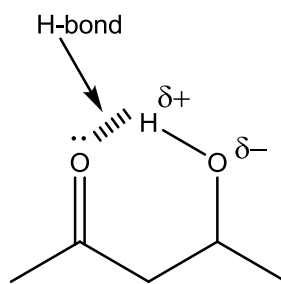
1 (a) (i) Functional group isomer

(ii)
$$K_c = \frac{[\text{CH}_3\text{C}(\text{OH})\text{CH}_2]}{[\text{CH}_3\text{COCH}_3]}$$

OR
$$K_c = \frac{[\text{enol}]}{[\text{propanone}]}$$

(iii) The small equilibrium constant indicates that the equilibrium position lies to the left hence at equilibrium, most of the species present is the propanone OR there is very little enol.

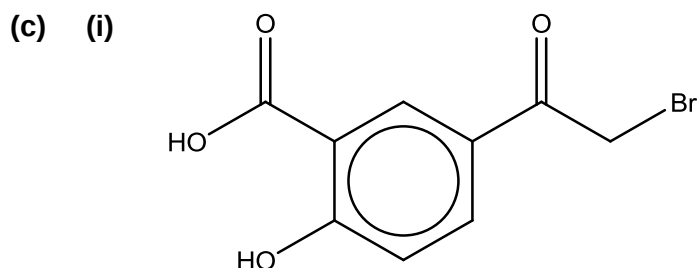
(iv) The large equilibrium constant indicates that at equilibrium, there is more of the compound B (enol) compared to compound A (diketone) present. This is because compound B (the enol) is stabilised by intramolecular hydrogen bonding.



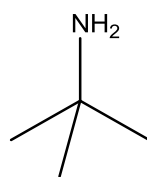
(b) (i) The acid acts as a catalyst.

(ii) HBr

(iii) There will be no change in the rate as the halogen is not involved in the rate determining step.



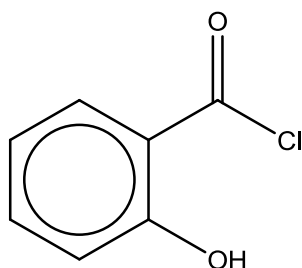
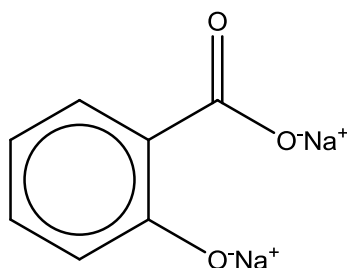
(ii)



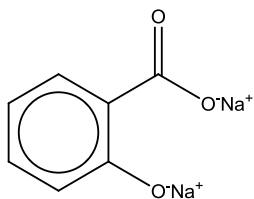
Step II : excess ethanolic , heat with reflux
Step III: LiAlH_4 in dry ether, room temperature

(d) (i) Product with Na metal:

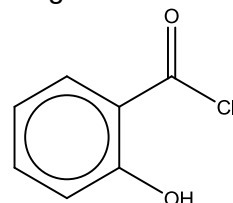
Product with PCl_5



(ii)



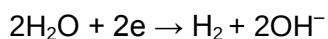
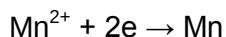
has a giant ionic structure held by strong electrostatic



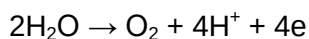
forces of interaction between oppositely charged ions while has simple molecular structure held by intermolecular hydrogen bonds. More energy is required to break the stronger electrostatic forces of interaction hence causing the ionic compound to have a larger boiling point.

- (iii) The product will hydrolyse more readily than chloroethane as the C atom on the acyl group is highly electron deficient due to the presence of 2 electronegative atoms Cl and O. This makes it more reactive and hence more susceptible to even weak nucleophile, H_2O . Chloroethane is less electron deficient as the carbon atom is bonded to only 1 electronegative atom, Cl. This makes it less reactive and hence a stronger nucleophile, OH^- is needed for the hydrolysis. Product formed is AgCl

2 (a) (i) Cathode:



(ii) Anode:



(iii)

$$\text{Amount of electrons needed to produce } 1 \times 10^6 \text{ g Mn} = \frac{1 \times 10^6 \text{ g}}{54.9} \times 2$$

$$= 3.643 \times 10^4 \text{ mol}$$

$$Q = It = nF$$

$$I \times (24 \times 60 \times 60) = (3.643 \times 10^4)(9.65 \times 10^4)$$

$$\text{Current needed, } I = 4.07 \times 10^4 \text{ A}$$

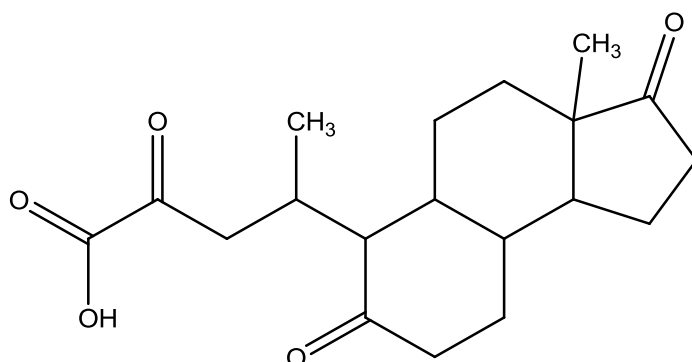
(iv) The brown solid is MnO_2 .

The Mn^{2+} in the solution reacts with the O_2 evolved at the anode to produce MnO_2 .

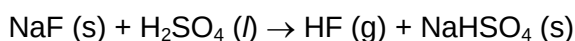
(b) (i) Purple KMnO_4 decolourises.

(ii) Ketone or carboxylic acid

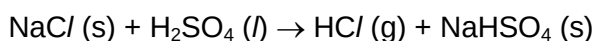
(iii)



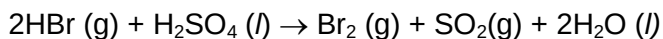
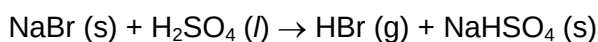
(c) NaF : steamy white fumes of HF evolved and HF will not be further oxidised



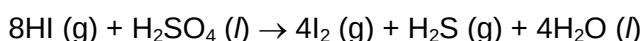
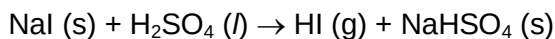
NaCl : steamy white fumes of HCl evolved and HCl will not be further oxidised



NaBr : steamy white fumes of HBr and some HBr will be further oxidised to orange-brown fumes of Br_2 evolved. Pungent SO_2 evolved.



I-(NaI): All the HI will be further oxidised. Violent fumes of I_2 and some steamy white fumes of HI evolved. Pungent H_2S evolved.



(d) (i) $\text{SO}_2 \text{ (g)} + \text{Cl}_2 \text{ (g)} \rightleftharpoons \text{SO}_2\text{Cl}_2 \text{ (g)}$

$$K_p = \frac{P_{\text{SO}_2\text{Cl}_2}}{P_{\text{SO}_2} P_{\text{Cl}_2}}$$

- (ii) When the volume of container is decreased, the pressure increases. By LCP, the equilibrium position shifts to the right to reduce the number of gaseous molecules and to reduce the pressure favouring the forward reaction. K_p has no change as temperature remains unchanged.

3 (a) (i) $K_{sp} = [Ca^{2+}][C_2O_4^{2-}]$

- (ii) Let the solubility of calcium oxalate be $s \text{ mol dm}^{-3}$

$$\begin{aligned} K_{sp} &= [Ca^{2+}][C_2O_4^{2-}] \\ 1.96 \times 10^{-8} &= [Ca^{2+}][C_2O_4^{2-}] = s^2 \\ s &= [C_2O_4^{2-}] = 1.40 \times 10^{-4} \text{ mol dm}^{-3} \end{aligned}$$

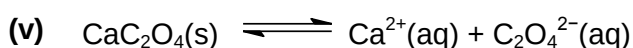
- (iii) Based on the data, the expected amount of calcium lost in the urine is 100 to 300 mg per day.

$$\begin{aligned} &\text{Taking the maximum of that range, } [Ca^{2+}] \text{ present in the urine} \\ &= (300 \times 10^{-3}) / 40.1 \\ &= 7.481 \times 10^{-3} \text{ mol dm}^{-3} \end{aligned}$$

$$\begin{aligned} &\text{In order for } CaC_2O_4 \text{ to precipitate, ionic product} = K_{sp}, \text{ hence} \\ 1.96 \times 10^{-8} &= 7.481 \times 10^{-3} \times [C_2O_4^{2-}] \\ [C_2O_4^{2-}] &= 2.62 \times 10^{-6} \text{ mol dm}^{-3} \end{aligned}$$

- (iv) Mass of oxalic acid that must be consumed
 $= (2.62 \times 10^{-6}) \times 90$
 $= 2.358 \times 10^{-4} \text{ g}$

$$\begin{aligned} &\text{Since spinach has 0.97\% oxalic acid by mass, the mass of spinach required} \\ &= 2.358 \times 10^{-4} \text{ g} / (0.0097) \\ &= 0.0243 \text{ g} \end{aligned}$$



Consumption of large amounts of water decreases the $[Ca^{2+}]$ and $[C_2O_4^{2-}]$.
 By Le Chatelier's Principle, the equilibrium position shifts right to increase $[Ca^{2+}]$ and $[C_2O_4^{2-}]$. Hence, the kidney stones can dissolve more readily.

- (b) (i) Entropy is a measure of the disorder in a system.
- (ii) The sign of ΔS for dialysis fluid is positive. During the hemodialysis process, the waste particles in the dialysis fluid is increased. There are more particles and hence more ways of arranging the particles. Entropy increases/system becomes more disordered and ΔS is positive.

- (c) (i) Volume of CO_2 produced by adult = $4/100 \times 500$
 $= 20.0 \text{ cm}^3$

$$pV = nRT$$

$$\text{No. of moles of CO}_2 \text{ produced} = \frac{1.01 \times 10^5 \times 20.0 \times 10^{-6}}{8.31 \times 310} = 7.84 \times 10^{-4} \text{ mol}$$

- (ii) The gas behaves like an ideal gas/ has negligible intermolecular forces of attraction.

This assumption is invalid because of the significant intermolecular forces of attraction in CO₂.

OR

Negligible volume of particles compared to the total volume of the gas/volume of the container. Or it is valid as the CO₂ molecules has negligible volume compared to volume of container.

- (iii) Amount of H₂SO₄ in 25 cm³ aliquot = amount of NaOH reacted / 2
 $= \left(\frac{10}{1000} \times 0.250 \right) / 2$
 $= 0.00125 \text{ mol}$

$$\text{Amount of H}_2\text{SO}_4 \text{ in } 250 \text{ cm}^3 = 0.00125 \text{ mol} \times 10$$

$$= 0.0125 \text{ mol}$$

$$\text{Amount of H}_2\text{SO}_4 \text{ used} = \frac{100}{1000} \times 0.250$$

$$= 0.0250 \text{ mol}$$

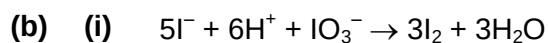
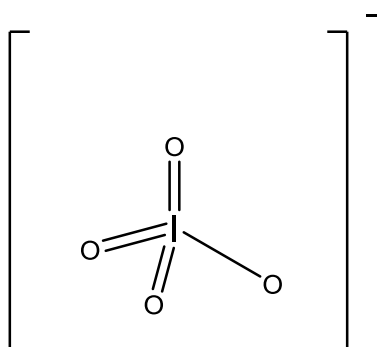
$$\text{Amount of H}_2\text{SO}_4 \text{ reacted with CaCO}_3 = 0.0250 - 0.0125$$

$$= 0.0125 \text{ mol}$$

$$\text{Amount of CO}_2 \text{ present in exhaled air} = 0.0125 \text{ mol}$$

- (iv) No. of breath = $0.0125 / 0.000784 = 15.94 = 16$

4 (a)



(ii) Amount of $\text{NaIO}_3 = \left(\frac{0.96}{198} \right) = 4.848 \times 10^{-3} \text{ mol}$

Amount of $\text{NaI} = \left(\frac{4.0}{150} \right) = 0.02667 \text{ mol}$

Since $5\text{I}^- \equiv \text{IO}_3^-$,

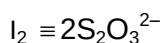
IO_3^- is the limiting reagent

Since $\text{IO}_3^- \equiv 3\text{I}_2$,

Total amount of iodine produced $= 4.848 \times 10^{-3} \times 3$

$$= 0.01455 \text{ mol} = 0.0145 \text{ mol}$$

(iii) Amount of $\text{S}_2\text{O}_3^{2-} = \frac{21.60}{1000}(0.1) = 0.002160 \text{ mol}$



Amount of iodine which reacted with $\text{S}_2\text{O}_3^{2-} = \frac{0.00216}{2} = 0.001080 \text{ mol}$

Amount of iodine which reacted with ascorbic acid $= 0.01455 - 0.001080$
 $= 0.01347$
 $= 0.0135 \text{ mol}$

(iv) Amount of ascorbic acid $= 0.01347 \text{ mol}$
 Mass of ascorbic acid $= 0.01347 \times 176 = 2.37\text{g}$

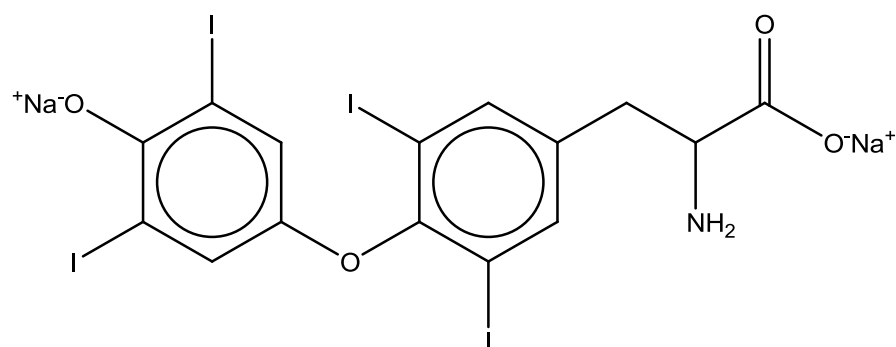
(c) (i) Tertiary structure refers to the overall 3-dimensional shape of a polypeptide chain involving folding or coiling of the chain. Folds are stabilized by interactions between R groups (side chain) of amino acids in the poly chain. There are 4 main types of R group interactions: H bonding, disulfide linkages, van der Waals' interactions and ionic bonds.

(ii) Asn-Ile-Phe-Glu-T4-Gln-Val-Asp-Ala-Gln-Pro-Leu-Arg-Pro-Cys-Glu-Leu-Gln-Arg

(iii) Dilute/ Aqueous HCl or H_2SO_4 or NaOH (aq) , prolonged heating with reflux

(iv) Halogenoarenes do not undergo nucleophilic substitution. This is due to the overlapping of p-orbitals of the iodine with p-orbitals of the benzene ring to form a delocalised π electron cloud. The delocalization of electrons shortens and strengthens the aryl-iodine/ C-I bond, making it too strong.

(v)



(i) $\text{Pt(s)} \mid \text{SO}_2(\text{g}) \mid \text{SO}_4^{2-}(\text{aq}) \parallel \text{I}_2(\text{g/aq}) / \text{I}_3^-(\text{aq}) \mid \text{I}^-(\text{aq}) \mid \text{Pt(s)}$

(ii) $\text{I}_2 + 2\text{e}^- \rightleftharpoons 2\text{I}^-$ $E^\ominus = +0.54 \text{ V}$ at standard conditions

As the concentration of I^- is 2.0 mol dm^{-3} , the system will try to decrease the concentration of I^- by shifting the equilibrium to the left. This causes the $E_{\text{I}_2/\text{I}^-}$ value to be less than $+0.54 \text{ V}$ hence resulting in the measured e.m.f having a smaller value.

(e) (i) Dry conditions are necessary as H_2O will react with the magnesium metal hence deactivating it.

(ii)	Steps	Reagents	Conditions
	1	$\text{CH}_3\text{CH}_2\text{I}/\text{Br}/\text{Cl}$, Mg Dry ether , trace amount of I_2 ,	Heat with reflux
	2	CH_2O , $\text{CH}_3\text{CH}_2\text{Mg I}/\text{Br}/\text{Cl}$ Dry ether ,	heat with reflux
	3	$\text{H}_2\text{SO}_4/\text{HCl}$ or H_3O^+ ,	room temperature

5	(a)	(i)	Comparing experiment 1 and 2, when $[\text{ClO}_2]$ increase by 2 times, keeping $[\text{OH}^-]$ constant, initial rate is increased by 4 times, order of reaction w.r.t. ClO_2 is 2. Comparing experiment 1 and 3, when $[\text{OH}^-]$ increases by 3 times, keeping $[\text{ClO}_2]$ constant, initial rate is increased by 3 times, order of reaction w.r.t OH^- is 1. OR <u>By substitution method :</u> Let the rate equation be : $\text{rate} = k[\text{ClO}_2]^x[\text{OH}^-]^y$ $\frac{\text{Experiment 2}}{\text{Experiment 1}} = \frac{\text{Rate}_2}{\text{Rate}_1} = \frac{k[\text{ClO}_2]_2^x [\text{OH}^-]_2^y}{k[\text{ClO}_2]_1^x [\text{OH}^-]_1^y}$ $\frac{2.48 \times 10^{-2}}{6.21 \times 10^{-3}} = \frac{k(0.1)^x (0.0216)^y}{k(0.05)^x (0.0216)^y}$ $\Rightarrow 4 = 2^x$ $\Rightarrow x = 2$ Order wrt $\text{ClO}_2 = 2$ $\frac{\text{Experiment 3}}{\text{Experiment 1}} = \frac{\text{Rate}_3}{\text{Rate}_1} = \frac{k[\text{ClO}_2]_3^x [\text{OH}^-]_3^y}{k[\text{ClO}_2]_1^x [\text{OH}^-]_1^y}$ $\frac{1.86 \times 10^{-2}}{6.21 \times 10^{-3}} = \frac{k(0.05)^x (0.0648)^y}{k(0.05)^x (0.0216)^y}$ $\Rightarrow 3 = 3^y$ $\Rightarrow y = 1$ Order wrt $\text{OH}^- = 1$
		(ii)	Rate = $k[\text{ClO}_2]^2[\text{OH}^-]$
		(iii)	Using experiment 1, $6.21 \times 10^{-3} = k(0.0500)^2(0.0216)$ $k = 115 \text{ mol}^{-2} \text{ dm}^6 \text{ s}^{-1}$
		(iv)	$0.145 = 115 [0.200]^2[\text{OH}^-]$ $[\text{OH}^-] = 0.0315 \text{ mol dm}^{-3}$ $\text{pOH} = -\log [\text{OH}^-]$ $= 1.50$ $\text{pH} = 12.5$
	(b)	(i)	The CFC molecules break down to form <u>chlorine radicals</u> , which initiate a chain reaction of destruction of ozone molecules or act as a <u>catalyst</u> .

			$\text{CF}_3\text{Cl} \xrightarrow{\text{uvlight}} \text{CF}_3\bullet + \text{Cl}\bullet$ The chlorine radicals act as homogeneous catalyst for the destruction of ozone.
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		(ii)	Fraction of particles 0 E_a Energy, E Fraction of particles with $E \geq E_a$ at high T Fraction of particles with $E \geq E_a$ at low T Effect of increasing the temperature : ➤ Average kinetic energy of the particles increases. ➤ There is an increase in the fraction of particles with energy equal to or greater than the activation energy, E_a. ➤ Frequency of effective collisions taking place in the reaction increases. ➤ Since rate of reaction is proportional to the frequency of effective collisions, rate of reaction increases.
		(iii)	Most of the high energy uv radiation is absorbed in the stratosphere hence there is little uv radiation in the lower atmosphere.

		(c)	P is an acid as it is readily soluble in aqueous sodium hydroxide. P reacts via nucleophilic substitution to form Q. Q has the functional group $\text{CH}_3\text{CH}(\text{OH})-$ or $\text{CH}_3\text{C}=\text{O}-$ Q reacts via mild oxidation to form a yellow ppt, CHI_3. Q reacts via elimination to form R, an alkene. R reacts via electrophilic addition to form the shown compound above. R: $\begin{array}{c} \text{H} \quad \text{H} \\ \quad \\ \text{H}_3\text{C}-\text{C}=\text{C}-\text{CH}_2\text{COOH} \end{array}$ Q: $\begin{array}{c} \text{H} \quad \text{H} \\ \quad \\ \text{H}_3\text{C}-\text{C}-\text{C}-\text{CH}_2\text{COOH} \\ \quad \\ \text{OH} \quad \text{H} \end{array}$
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		$ \begin{array}{c} \text{H} \quad \text{H} \\ \quad \\ \text{H}_3\text{C}-\text{C}-\text{C}-\text{CH}_2\text{COOH} \\ \quad \\ \text{Cl} \quad \text{H} \end{array} $ P:
	(d)	White ppt of AgCl is formed when NaCl reacts with AgNO₃ $\text{AgX (s)} \rightleftharpoons \text{Ag}^+ \text{(aq)} + \text{X}^- \text{(aq)} \quad \text{----- (1)}$ OR any specific eqn for AgCl and AgI When dilute NH₃ is added, Ag⁺ will react with it to form a stable diamminesilver(I) complex Ag(NH₃)₂⁺. $\text{Ag}^+ \text{(aq)} + 2\text{NH}_3 \text{(aq)} \rightleftharpoons [\text{Ag}(\text{NH}_3)_2]^+ \text{(aq)}$ This lowers [Ag⁺(aq)] and causes the equilibrium (1) to shift to the right. Hence AgX becomes more soluble. Ionic product [Ag⁺][Cl⁻] is <i>lower</i> than the K_{sp} (AgCl). AgCl ppt is <i>soluble</i> in dilute NH₃ (aq) hence no ppt is formed. Ionic product [Ag⁺][I⁻] is <i>larger</i> than the K_{sp} (AgI). AgI ppt is <i>insoluble</i> in dilute NH₃ (aq) hence the yellow ppt is formed. as the K_{sp} of AgCl > AgI